Band-1 Computational Experiment: BSD on one curve — L-value / rank pipeline (Cremona 11a1)

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: BSD on one concrete elliptic curve

Honesty stamp. This note documents an **incremental** L-value / rank pipeline for **one** elliptic curve over $\mathbb{Q}$. It does **not** claim the Birch–Swinnerton-Dyer conjecture (BSD) is proved, for this curve or in general. It does **not** claim a Clay Millennium Problem resolution. It does **not** stamp a finished Theory of Everything. Every numeric claim is labeled **TABLE LOOKUP** (LMFDB / Cremona) or **BOX COMPUTATION** (pure Python on this machine). Sage / PARI were **not** available.

Band callout. Full BSD (leading-coefficient identity with |Sha|, periods, regulator, Tamagawa) remains **open in general** (Millennium). For this specific curve of analytic rank 0, Kolyvagin / Gross–Zagier / modularity machinery already implies much of the rank part in the literature; **this lab does not re-prove that**.

Abstract

Curve lock: Cremona `11a1` = LMFDB `11.a2`,

$$y^2 + y = x^3 - x^2 - 10x - 20, \qquad [a_1, a_2, a_3, a_4, a_6] = [0, -1, 1, -10, -20], \qquad N = 11.$$

Model for $X_0(11)$; smallest conductor of any elliptic curve over $\mathbb{Q}$.

Ranks compared (tables): algebraic rank $r = 0$ and analytic rank $r_{\text{an}} = 0$ **agree**. Box role: verify Frobenius traces a_p / Dirichlet coefficients a_n against the modular form `11.2.a.a`, check torsion points on the model, and report crude Dirichlet partial sums as **rank-0 sanity indicators only** (not Dokchitser L-values).

Wall clock: ≈ 12 s for a_p at all primes $p \leq 20000$ + multiplicative a_n.

Problem statement (increment, not full BSD)

BSD predicts, for $E/\mathbb{Q}$,

$$\text{ord}_{s=1} L(E, s) = \text{rank}\, E(\mathbb{Q}),$$

and an identity for the leading coefficient involving |Sha|, Ω_E, Reg, Tamagawa numbers, and torsion.
Tonight's increment:

1. Lock one standard curve.
2. Compute / verify a_p on the box.
3. Extend to a_n; cross-check the published q-expansion.
4. Compare algebraic vs analytic rank from **tables**.
5. Record crude partial sums $S_M(1) = \sum_{n \le M} a_n n^{-1}$ as indicators.
6. Ledger the published BSD special-value arithmetic (floats only).

We do **not** run Dokchitser / approximate functional equation on this box, and we do **not** certify $|\mathrm{Sha}| = 1$.

Curve lock

Field	Value	Status
Cremona	`11a1`	standard
LMFDB	`11.a2`	https://www.lmfdb.org/EllipticCurve/Q/11/a/2
Conductor N	11	tables
Minimal model	$y^2 + y = x^3 - x^2 - 10x - 20$	tables
Discriminant Δ	$-11^5 = -161051$	tables
CM	none	tables
Mordell–Weil	$E(\mathbb{Q}) \cong \mathbb{Z}/5\mathbb{Z}$	TABLE
Algebraic rank r	0	TABLE
Analytic rank r_{an}	0	TABLE
Torsion generators (affine)	$(5, 5)$ order 5	TABLE; box-checked on model
Modular form	`11.2.a.a`	LMFDB
Bad prime	11, split multiplicative, $c_{11} = 5$, local root number -1	TABLE

What BSD asserts (reminder only)

$$\frac{L^{(r)}(E,1)}{r!} \stackrel{?}{=} \frac{|\mathrm{Sha}(E/\mathbb{Q})| \cdot \Omega_E \cdot \mathrm{Reg}(E/\mathbb{Q}) \cdot \prod_p c_p}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}.$$

For rank 0: $L(E,1)$ on the left, $\mathrm{Reg} = 1$.

Literature BSD ledger for 11a1 (TABLE LOOKUP)

Values from LMFDB (accessed 2026-09-11) / Cremona. **Not** recomputed by Dokchitser here.

Quantity	Literature value	Role
r_{an}	0	order of vanishing at $s = 1$
$L(E,1)$	$\approx 0.2538418608559107$	special value
Real period Ω_E	$\approx 1.2692093042795534$	Néron period
Regulator Reg	1	rank 0
$\prod c_p$	5	Tamagawa at $p = 11$
$\lvert E(\mathbb{Q})_{\mathrm{tors}}\rvert$	5	torsion
$\mathrm{Sha}_{\mathrm{an}}$	1 (exact, analytic)	analytic Sha

Numerical consistency check (literature arithmetic only):

$$\frac{1 \cdot 1.2692093042795534 \cdot 1 \cdot 5}{5^2} = 0.2538418608559107,$$

matching $L(E,1)$ to the printed digits (box abs error 0 at float precision used). **This is a ledger identity among published floats, not a proof of BSD and not a proof that** $|\text{Sha}| = 1$**.**

Box computations (this lab)

Environment. `sage` / `gp` (PARI) **not installed**. Pure Python point-counts over $\mathbb{F}_p$ and multiplicative extension of a_n.

A. Frobenius traces a_p — BOX COMPUTATION

For good $p \neq 11$,

$$a_p = p + 1 - |E(\mathbb{F}_p)|$$

(including the point at infinity). For $p = 11$ (split multiplicative), $a_{11} = +1$ (newform / local convention, matching LMFDB `11.2.a.a`).

Result (excerpt, all $p \leq 20000$ **computed on box; Hasse bound verified for good** p**):**

p	2	3	5	7	11	13	17	19	23	29	31	37	41	43	47
a_p	-2	-1	1	-2	1	4	-2	0	-1	0	7	3	-8	-6	8

Status: box-proved finite arithmetic (point enumeration), consistent with the published q-expansion of `11.2.a.a`.

B. Dirichlet coefficients a_n — BOX COMPUTATION

Weight-2 Euler recurrence

$$a_{p^k} = a_p\, a_{p^{k-1}} - \mathbf{1}_{p\nmid N}\, p\, a_{p^{k-2}},$$

plus multiplicativity, filled through $n \leq 20000$. Comparison with LMFDB through $O(q^{20})$:

$$(a_n)_{n=1}^{19} = (1, -2, -1, 2, 1, 2, -2, 0, -2, -2, 1, -2, 4, 4, -1, -4, -2, 4, 0).$$

Exact match to LMFDB `11.2.a.a`. **Status: box-proved** relative to the a_p table and the standard newform recurrence.

C. Crude Dirichlet partial sums — numerical only; not L-values

$$S_M(s) = \sum_{n \leq M} a_n n^{-s}.$$

Object	Value	Label
$S_{1000}(1)$	≈ 0.2618	numerical partial sum only
$S_{5000}(1)$	≈ 0.3144	numerical

Object	Value	Label
$S_{20000}(1)$	≈ 0.2095	numerical
$S_{5000}(1.5)$	≈ 0.4079	numerical
$S_{5000}(2)$	≈ 0.5461	numerical
Literature $L(E,1)$	≈ 0.25384186085591	**TABLE**

Honesty. Without the completed L-function / approximate functional equation (Dokchitser et al.), truncated Dirichlet series **do not** equal $L(E,1)$. At the central point convergence is conditional and oscillatory ($S_M(1)$ wanders in a $O(0.1\text{–}0.3)$ band here). Prefer the literature Dokchitser value. The only soft analytic-rank **indicator** claimed on box is that $S_M(1)$ does **not** collapse toward 0 for these M — consistent with $r_{\text{an}} = 0$, **not** a certificate.

D. Torsion points on the model — BOX COMPUTATION

Affine integral points $(5,5)$, $(5,-6)$, $(16,60)$, $(16,-61)$ satisfy the Weierstrass equation over $\mathbb{Z}$. **Status: elementary verification.** That they generate $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/5\mathbb{Z}$ is **TABLE** (not re-proved by 2-descent / saturation tonight).

Ranks compared

Rank	Value	Source
Algebraic $r = \operatorname{rank} E(\mathbb{Q})$	0	**TABLE** (LMFDB / Cremona)
Analytic $r_{\text{an}} = \operatorname{ord}_{s=1} L(E,s)$	0	**TABLE** (LMFDB)
Agreement	**yes**	tables
Box contribution	a_p/a_n match + partial-sum sanity	**BOX**

Pipeline diagram (documented increment)

```
Weierstrass model (11a1)
        |
        v
  #E(F_p)  -->  a_p = p+1-#E(F_p)      [BOX]
        |
        v
  Euler recurrence --> a_n              [BOX]
        |
        +---- cross-check q-exp 11.2.a.a  [BOX vs TABLE]
        |
        v
  S_M(s) = sum a_n n^{-s}               [BOX numerical indicator]
        |
        +---- compare to L(E,1), r_an   [TABLE]
        |
        v
```

```
rank_alg (TABLE)  ?=?  rank_an (TABLE)
      |
      v
BSD product ledger (TABLE floats)    [NOT a proof]
```

Missing for full BSD certification on box: Dokchitser $L^{(r)}(E,1)$, Néron period integral, canonical-height regulator (trivial here), Tamagawa computation from Kodaira symbols, and a proof that $|\text{Sha}| = 1$.

Unit checks

Check	Result
$a_n[1..19]$ match LMFDB	PASS
Torsion sample points on curve	PASS
BSD table-float product vs $L(E,1)$	PASS (abs err 0 at used precision)
Hasse $\lvert a_p\rvert \le 2\sqrt{p}$ for good $p \le 20000$	PASS

Artifacts

- Code / data: `/workspace/experiments/band1/bsd_one_curve/`
- `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_bsd_one_curve.log`
- This paper: `2026-09-11_Band1_Computational_Experiment_bsd_one_curve.md` (+ PDF)

References (ledger)

- LMFDB, Elliptic curve 11.a2 (Cremona 11a1): https://www.lmfdb.org/EllipticCurve/Q/11/a/2
- J.E. Cremona, *Algorithms for Modular Elliptic Curves*; online elliptic-curve tables.
- Modular form 11.2.a.a (LMFDB).
- Birch–Swinnerton-Dyer conjecture (Clay Millennium problem statement).
- Dokchitser, Computing special values of L-functions (for what a full pipeline needs).

Bottom line

Curve: Cremona `11a1` / LMFDB `11.a2`.
Ranks compared: algebraic 0 (TABLE) = analytic 0 (TABLE).
Box: a_p/a_n verified vs `11.2.a.a`; torsion points on model; partial sums logged.
Full BSD: not claimed.